

TRE HUMPHRIES

Hardware / Software Generalist | Controls | Embedded Systems | Industrial Automation | Internal Tools

Remote-first | Open to hybrid NYC/NJ | Open to travel for high-leverage field work | (573) 933-0405 | trehumphries16@gmail.com |
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SUMMARY

Hardware/software generalist who bridges controls, embedded systems, industrial automation, and internal software tools. Strong fit for hard-tech startups that need someone who can debug real machines, understand field failures, build practical software around operations, and translate messy technical problems into systems that work.

Trusted in small-team environments to move across controls logic, PLCs, VFDs, edge data, field troubleshooting, React/Python tooling, REST APIs, dashboards, and business-critical workflows without waiting for perfect structure.

EXPERIENCE

Senior Engineering & Technology Developer - GAF Roads / Standard Industries

Feb 2025 - Jan 2026

Remote / NJ | Project ending due to initiative shutdown

- First dedicated controls/systems engineer hired to stabilize an early EV sprayer platform during a critical reliability and production-readiness phase.
- Restored field performance to production readiness after rapid onsite debugging trips to Phoenix and Toronto, working across controls logic, mechanical behavior, edge data, and operator workflows.
- Reworked and debugged Codesys-based embedded controls, edge data architecture, OTA readiness, and cloud integration paths for a hardware/software platform operating in real field conditions.
- Served as technical bridge between mechanical engineering, field operations, and digital/software teams, translating field failures into roadmap decisions and actionable engineering fixes.

Automation Engineer - Mainstream Fluid & Air

May 2024 - Feb 2025

Berkeley Heights, NJ

- Functioned as the internal tech department for a 30-person OEM, owning high-leverage software, controls, automation, and workflow problems without a dedicated engineering software team.
- Designed and built internal quoting, fan-selection, dashboarding, and process automation tools that converted spreadsheet-heavy workflows into repeatable software-backed systems.
- Built full-stack internal tools using React, Python/Flask, PostgreSQL, REST APIs, authentication/roles, and engineering calculation logic tied to real product selection constraints.
- Reduced reliance on tribal knowledge between sales, engineering, and operations by turning product rules, pricing logic, and handoff assumptions into maintainable internal systems.

Controls Engineer - Mainstream Fluid & Air

Aug 2022 - May 2024

Berkeley Heights, NJ

- R&D and customer-facing controls lead for custom industrial fan array systems used in commercial HVAC applications.
- Designed and deployed PLC, HMI, VFD, BACnet, and Modbus systems for mission-critical commercial installations.
- Supported startup, commissioning, quality, and field troubleshooting by connecting control logic intent to real equipment behavior.

Controls Engineer - Innerspec Technologies

Jan 2021 - Jul 2022

Forest, VA

- Delivered industrial automation systems using Allen-Bradley, Siemens, Beckhoff PLCs, motion systems, and field-integrated ultrasonic inspection equipment.
- Built experience debugging hardware/software/control interactions under real equipment constraints.

SELECTED PROJECT

Internal Fan Selection & Quoting Platform - Mainstream Fluid & Air

Built a full-stack internal platform that replaced spreadsheet-heavy fan selection and quoting workflows with a more reliable software-backed process. Implemented the React frontend, Flask backend, PostgreSQL database, authentication/roles, pricing logic, engineering calculations, product-selection workflows, and deployment around a Windows-only vendor performance DLL.

The platform turned a tribal-knowledge workflow across sales and engineering into a clearer, repeatable system with better handoffs and less dependence on individual memory.

SKILLS

Controls / Industrial: PLCs, HMIs, Codesys, Allen-Bradley, Siemens, Beckhoff, VFDs, BACnet, Modbus, EtherCAT, motion systems, commissioning, startup, field troubleshooting

Software / Tools: Python, JavaScript, React, Flask, REST APIs, PostgreSQL, dashboards, internal tools, authentication/roles, process automation, DevOps/CI practices

Startup Strengths: hardware/software bridge, ambiguous technical problem solving, field-to-engineering translation, internal tooling, cross-functional execution

EDUCATION

Purdue University - B.S. Mechanical Engineering Technology | West Lafayette, IN | 2016 - 2020

Interests: Music production, philosophy, horticulture, skateboarding | **Languages:** English native, Portuguese conversational, Spanish conversational